

NLP Course — Exam Topics by Week

01 - Intro

22.03

02 - Markov LM

Language Modeling, prompt, Unigram model

Bigram model, Markov, Higher-order Markov Models, MLE, smoothing, Back-off Models, Linear Interpolation, generalization, Pereira's solution, perplexity

12.04

03 - Classification and log linear models

19.04

Classification, MLE, Naive Bayes, bag of words, discriminative approach, log linear models, sentiment analysis (including negation)

04 - Sequence Methods (HMM)

Part of speech

HMM, Viterbi Algorithm, pseudowords

26.04

05 - Sequence Methods (Discriminative Models)

MEMM, CRF

06 - RNNs

Negation in sentiment analysis, context length, RNN

RNN, sentiment analysis, generalization, backpropagation, vanishing and exploding gradients, LSTM

10.05

07 - Syntax

Grammar, words

Words, morphology, segmentation and tokenization, clauses, noun phrases, syntactic heads, dependencies, information from syntactic structure, a proxy to semantics

17.05

08 - Word Embeddings

Distributional hypothesis, count-based methods, dimensionality reduction, prediction based models, skip-grams

09 - Graph-based dependency parsing

24.05

Dependency parsing, graph based parsing, MST parser, neural MST parsing

10 - Machine Translation

31.05

Ambiguity resolution, linguistic differences, lexical gaps, Vauquois triangle, direct transfer, statistical methods, neural MT, attention & transformers

Attention & transformers, MT evaluation, BLEU Score, modified N-Gram Precision

07.06

11 - Pretraining and Contextualized Embeddings

Contextualized word embeddings, pretraining, fine tuning

BERT, zero shot, few shot, decoder only models, LM evaluation, LMJ

14.06

12 - Post Training

Instruct LM, post training

RLHF, agents

21.06

13 - Tokenization

Character level, subword, BPE, morphology-aware

14 - Semantics

Information Extraction, Named Entity Recognition, Co-reference Resolution, Relation Extraction, Distant Supervision, Path Features, Schematized Algorithm for Distant Supervision, SRL